

Cosmology Research Programme: Investigations, Results and Pending Tests

A five-page technical summary of the PLAMB/FR, supernova, SPARC, SU(2), Quaia, HHT, SkyMap and BBN work

Prepared for the PLAMB research programme

17 July 2026 - current evidence summary

Research status

The programme now has reproducible tests across standard candles, galaxy rotation curves, sky anisotropy and primordial abundances. Several targeted patterns remain interesting, but no stream currently passes all promotion gates.

The governing claim boundaries are: SPARC subset wins, not a full-sample win; Quaia is a follow-up candidate only; FR/LINX is a standard expanding-background proxy; and the SU(2)-CMB thermal benchmark is separate from the late-time SU2/SU2R fluid.

Abstract

The research began with phenomenological tests of whether redshift-distance relations, light propagation or clock-rate effects could imitate part of the standard cosmological signal. It expanded into a SPARC optical-depth representation of the radial acceleration relation, SU2/SU2R background fits, Quaia angular scans, SkyMap and Hilbert-Huang Transform (HHT) residual searches, and finally Big Bang nucleosynthesis (BBN) calculations using LINX. The strongest current contributions are methodological: full-covariance likelihoods, hierarchical calibration offsets, preregistered redshift frames, mock-catalogue look-elsewhere corrections, non-centred posterior sampling, injection-recovery calibration, immutable result manifests and independent numerical validation. Positive local patterns exist, but calibration, spatial clustering, predictive stress, sampler divergences and primordial-abundance incompatibilities prevent discovery language. The next phase should emphasise independent catalogues, externally justified priors, matched likelihoods and model-specific first-principles derivations.

Keywords: PLAMB; FR cosmology; supernovae; SPARC; SU(2); Quaia; HHT; BBN; LINX; reproducibility.

1. Supernovae, Calibration and Redshift Frames

Scope. The standard-candle programme uses Pantheon+SH0ES, DES-SN5YR/Dovekie and Union3.1 distance moduli with their published covariance or precision matrices. The principal code is `run_joint_rawmu_hierarchical_offset_likelihood.py`, supported by `plamb_rawmu_model_utils.py` and the July 16 no-loss FR/LCDM follow-up programmes. The promoted diagnostic is the hierarchical raw-MU likelihood, not a free per-catalogue shift.

Core hierarchical and multiplicative-frame definitions

$$\begin{aligned} \mu_i &= 5 \log_{10}[(c/H_0) z_i (1 + \beta z_i)] + 25 + \delta_d(i) + \delta_s(i) \\ \delta_d &= \text{Normal}(0, \sigma_{\text{dataset}}^2) \\ \delta_s &= \text{Normal}(0, \sigma_{\text{survey}}^2) \\ 1 + z_{\alpha} &= (1 + z_{\text{HEL}})[(1 + z_{\text{HD}})/(1 + z_{\text{HEL}})]^{\alpha} \end{aligned}$$

Table 1. Supernova tests completed through 16 July 2026

Completed test	Principal readout	Standing interpretation
Joint hierarchy	N = 3,422; moderate dataset+survey row β = 0.521314 at $H_0 = 67.5$	β remains close to 0.5, but the improvement is calibration-mediated.
Offset prior audit	Pantheon+ HD dataset pull about -2.71 sigma under moderate priors; >5 sigma under tight priors	Replace generic grids with release-based calibration budgets.
Preregistered frames	Pantheon+: zHD primary; DES: zHD primary; zHEL/zCMB controls	zHD is the default external-flow readout.
Frame crossing	Exact-log AIC preference disappears by α = 0.4739 (Pantheon+) and 0.7779 (DES)	The earlier heliocentric preference is redshift-frame sensitive.
FR no-loss ladder	HD FR beats free- Ω_m LCDM by base BIC in 4/4 budgets, posterior objective in 3/4	Favourable but not robust to calibration pulls or catalogue/redshift splits.
Follow-up gates	Union3.1-only and $z \geq 0.5$: 0/4 BIC wins and 0/4 posterior wins	High-redshift and independent-catalogue support is presently absent.

1.1 Current conclusion

The raw-MU branch remains a useful calibration diagnostic and an economical low-to-mid-redshift phenomenology. It is not yet a cosmological detection. Promotion requires externally documented dataset and survey calibration budgets, stable preference in leave-one-dataset-out and high-redshift tests, and a same-nuisance comparison with Lambda-CDM. zHD remains the headline frame; zHEL and zCMB remain controls unless a distinct physical model preregisters another choice.

2. Galaxy Acceleration and Angular Structure

2.1 SPARC optical-depth branch

Optical-depth representation of the radial acceleration relation

$$\begin{aligned} \tau &= [g_{\text{bar}}/(\kappa c H_0/(2\pi))]^p \\ \nu &= [1 - \exp(-\tau)]^{(-1)} \\ g_{\text{obs}} &= \nu g_{\text{bar}} \end{aligned}$$

The filtered all-Q2 family contains 157 galaxies and 3,001 rotation points after removing six preregistered persistent negative high-leverage systems. The earlier random-walk posterior placed p near one half. A separately registered four-chain non-centred NUTS run then achieved excellent rank diagnostics (maximum $\hat{R}$ 1.00259 for RAR and 1.00283 for PLAMB), but failed the locked zero-divergence criterion: 3 RAR and 6 PLAMB divergences. The in-sample PLAMB estimate was $p = 0.51892 \pm 0.00494$. These values improve numerical confidence but do not replace grouped predictive validation.

SPARC gate

Subset wins, not a full-sample win. The all-Q2, low-acceleration outer-point and inclination posterior-predictive gates remain explicit failures, and the new NUTS fit does not retroactively validate prior-redrawn stress scores.

2.2 Quaia/SU2 angular tests

Controlled angular residual model: dipole plus survey templates

$$r_i = a + d \cdot n_i + \sum_k \gamma_k T_{ik} + \epsilon_i$$

A redshift-window candidate near $z = 1.0\text{--}1.5$ survived several object-level checks, but the full all-window look-elsewhere probability was ordinary (p_{global} about 0.43 for the original strongest row). After catalogue-quality matching the targeted family had $p = 1$. Spatial-block mocks gave family $p = 0.0135, 0.0923$ and 0.2326 for 8, 12 and 16 degree cells, so the mode fails the requirement to remain below 1% across cluster scales. The locked direction has template $R^2 = 0.1636$ and variance-inflation factor 1.196, with strongest correlation -0.4044 to a Gaia scan-angle template.

Injection-recovery exposed an important refinement: at the observed 1x amplitude, strict physical recovery after scan-law and colour control was 0.2833, while the final BIC gate accepted only 0.0011; 0.2823 of trials were physically recovered and rejected only by BIC. The current template gate is therefore under-powered and collinear as well as non-confirmatory. The correct status is targeted follow-up candidate only, with no SU2 detection or headline anisotropy claim.

2.3 SkyMap and HHT archive

The historical SkyMap/Quaia and CMB-supernova HHT work has been curated with code, extracted reports and representative figures. HHT identified intrinsic-mode-function structure, but shuffled-order controls were ordinary: extended SU2-minus-Lambda-CDM maximum- and summed-energy upper-tail probabilities were 0.9703 and 0.9604. Archive this as method development and candidate-feature evidence; a new preregistered probe-separated analysis is required.

3. Primordial-Abundance Tests

3.1 FR/LINX validation

The FR investigation was translated into a standard expanding-background LINX clock/radiation proxy and scanned over 30,503 immutable key-network states. A preregistered 92-evaluation validation matrix then tested $\text{rtol} = 10^{-7}$, $\text{atol} = 10^{-11}$, 300/600 interpolation samples, key/small/full PRIMAT 2023 networks, updated neutron-lifetime states, the three main deuterium-sensitive rates and five omitted Li/Be channels. All 92 evaluations completed with no missing or failed rows.

Numerical and local deuterium-theory readout

$$\begin{aligned} \max |\Delta(10^5 D/H)| &= 5.51 \times 10^{-4} \\ \max |\Delta Y_p| &= 6.53 \times 10^{-5} \\ \max |\Delta(10^{10} \text{Li}/H)| &= 1.71 \times 10^{-3} \\ \sigma_{D, \text{theory}} &= 0.02608 \text{ at the strict full-network baseline} \end{aligned}$$

Table 2. Publication-grade FR/LINX validation rows

Strict full-network state	$10^5 D/H$	Y_p	$10^{10} \text{Li}/H$	Registered χ^2
Baseline	2.434914	0.2469325	5.637694	295.6 (reference scale)
Best registered direct-Li	2.585689	0.248639	4.647982	210.57

There are zero registered and zero updated-current simultaneous D+He+Li two-standard-deviation passes. Full-minus-key corrections are at most 0.000649 in D, 1.48×10^{-5} in helium and 0.0580 in lithium. The largest omitted-channel lithium response, Be7daap, is 0.06079 at baseline, far below the residual excess. The main statistical correction is instead the D theory budget: 0.02608 is about 90% of the registered observational error and must be marginalised jointly with rate pulls.

3.2 Literature-defined SU(2)-CMB bridge

High-temperature SU(2)-CMB benchmark

$$\begin{aligned} n_{\text{CMB}} &= 4^{(-1/3)} = 0.629960525 \\ H_{\text{SU2CMB}}/H_{\text{SBBN}} &= 1.25326 \text{ at } T_{\text{gamma}} = 1 \text{ MeV} \\ N_{\text{eff, SU2CMB}} &= 20.373 \text{ (LINX diagnostic)} \end{aligned}$$

The separately registered 62-point bridge propagates the published high-temperature SU(2)-CMB equation of state through LINX while retaining standard weak and nuclear microphysics. At fixed reference baryon density it predicts $D/H = 0.64441 \times 10^{-5}$, $Y_p = 0.3165$ and $\text{Li}/H = 26.476 \times 10^{-10}$. The best D+He row at eta factor 0.48 still has $\chi^2_{D+He} = 501.41$; the best row inside the published baryon prior has total $\chi^2 = 3602.7$. No abundance pass occurs. This is not the repository's late-time SU2/SU2R fluid and supplies no first-principles gauge-matter changes to masses, binding energies, Q-values or rates.

4. Pending Tests and Research Path

The next programme should be narrower, preregistered and designed around independent falsification. The following order gives the greatest gain in research standing per unit computation.

1. Quaia cross-catalogue validation

Repeat the $z = 1.0-1.5$ search in an independent quasar catalogue with mask, windows, latitude cuts and spatial-block null fixed in advance. Promotion requires a preregistered global p across sensible block scales, stable amplitude/direction, and a coherent $l = 1-3$ pattern.

2. Injection-calibrated angular likelihood

Replace the fixed $\Delta\text{BIC} < -10$ gate with a power-calibrated rule. Jointly model dust, depth, Gaia scanning geometry, WISE colour and catalogue-quality templates. The SU2 scalar-residual relation must beat template-only prediction on held-out sky regions.

3. Supernova calibration-budget hierarchy

Derive dataset and survey priors from release covariance and documented zero-point budgets. Keep zHD primary. Repeat same-nuisance FR versus Lambda-CDM comparisons with survey, dataset and high- z hold-outs; a robust result must stay within the external budget.

4. SPARC divergence and prediction gate

Retune or reparameterise both non-centred NUTS models to zero divergences, then run grouped galaxy-held-out prediction. Full-sample promotion requires the all-Q2, low-acceleration outer-point and inclination scores to reverse their gating failures under the locked sample and priors.

5. BBN hierarchy and first-principles models

Fit baryon density, neutron lifetime, key rate pulls and stellar lithium depletion with matched CMB/abundance likelihoods, then replicate independently. For FR, derive non-expanding $T(t)$, $\rho(T)$, $c(T)$, masses, Q_{np} , binding energies and rates. Keep SU(2)-CMB separate without an action-level bridge.

6. HHT/SkyMap confirmatory design

Define probe-separated residuals, frequency bands and sky masks before looking at the data; use phase-randomised and order-preserving surrogates, global correction and held-out redshift ranges. The existing archive is a hypothesis generator, not confirmatory evidence.

4.1 Reproducibility and evidence

Code, small results, reports, environment locks and manifests are curated at github.com/cliveboyd/Cosmology under `github_export/`. Large inputs and chains remain local but are checksum-referenced. Future papers should cite commit hashes, preregistrations, programme versions, dataset releases and claim gates.

Evidence index: [1] raw-MU hierarchy/frame audit (14 Jul); [2] SPARC full NUTS (16 Jul); [3] Quaia spatial-block audit (16 Jul); [4] FR/LINX matrix; [5] SU(2)-CMB bridge (17 Jul). Full paths: `github_export/results/README.md`.

Conclusion

The programme now has a defensible map of survival and failure. Highest-value work is independent Quaia validation, an externally budgeted supernova hierarchy, zero-divergence held-out SPARC prediction, and matched or first-principles BBN tests.